

Title: Provide Fuels Laboratory
Location: MCAS Miramar, CA

ATTACHMENT D

TECHNICAL SCOPE OF WORK

1. GENERAL

1.1. GENERAL DESCRIPTION

Demolition and construct/renovate the Fuel Laboratory Facility 9647 (CLIN 0001). The existing structure is in relatively good condition, therefore the building will be stripped down to its structural elements and renovated per the requirements indicated herein.

Design and construction of improvements will be in accordance with the latest version of the Marine Corps Air Station (MCAS) Miramar Requirements, Specific Facilities Design / Construction Requirements. See Attachment H for Rev 5, November 2022. Confirm current version with MCAS Miramar prior to design.

1.2. PHASING

The Contractor shall phase their work, in coordination with the installation, to minimize disruption to existing fuels operations. A phasing plan shall be submitted to the COR for review and approval. A written notice must be provided to the COR not less than 10 days in advance of any disruptions to fueling operations. The Contractor will establish free work zones in coordination with airfield operations prior to construction.

To allow Miramar personnel to continue daily work operations during the renovation and construction process, Miramar will be providing a mobile lab for their personnel. The contractor shall provide a tie-it to a fiber optic line with internet access and a storage/Conex box for personnel equipment until the laboratory renovation is complete and facility 9647 is occupiable. Miramar will determine the location of the storage/Conex box and will provide a map.

Contractor shall coordinate the haul route and laydown area with the installation. All equipment and materials will be stored within the laydown area for the duration of the contract unless approved in advance by the Contracting Officer's Representative (COR) and airfield manager. Construction office trailers will be located in the laydown area. The Contractor is responsible for maintaining and securing the lay down areas at all times. Low-profile barriers are required and must be approved by the airfield manager.

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2. ENVIRONMENTAL COMPLIANCE AND SUSTAINABILITY

2.1. General Requirements

This project consists of demolition and construct/renovate the Fuel Laboratory Facility 9647 (CLIN 0001). The existing structure is in relatively good condition, therefore the building will be completely renovated. Environmental considerations of this project includes demolition with hazardous building materials survey and abatement. Prevent environmental pollution and damage that may occur as a result of construction operations. The environmental resources within the project boundaries and those affected outside the limits of permanent work must be protected during the entire duration of this contract.

Comply with all applicable federal, state, and local environmental laws and regulations. Any delays resulting from failure to comply with environmental laws and regulations will be the Contractor's responsibility. The contractor is required to obtain all necessary federal, state, and local permits required for this project and review federal and state regulations and pay any required fees.

2.1.1. Sustainability and Energy Efficiency Policy

The Sustainability goal for this project is to reduce the amount of construction-related debris that is deposited in local landfills. Materials that are recyclable, e.g., concrete, asphalt, aluminum, copper, brass, steel, plastic, cardboard, glass, etc., shall be recycled to minimize the impact on local landfills. In accordance with Unified Facilities Guide Specification (UFGS), 01 74 19- A minimum of 60 percent by weight of total project solid waste shall be diverted from the landfill. The Contractor shall be responsible for the transportation and disposal of recyclable materials outside the limits of the government property.

2.2. Environmental Protection Plan

Prior to commencing construction/demolition activities or delivery of materials to the site, the Contractor shall submit an Environmental Protection Plan prepared in accordance with UFGS 01 57 19, Temporary Environmental Controls, for review and approval by the Contracting Officer. The Contractor shall maintain a current version of the Environmental Protection Plan on site for review by interested parties and implement any additional requirements in complying with applicable federal, state, and local environmental protection laws and regulations.

Copies of all environmental permits, permit application packages, approvals to construct, notifications, certifications, reports, and termination documents shall be attached, as an appendix, to the Environmental Protection Plan.

The Contracting Officer will notify the Contractor in writing of any observed noncompliance with federal, state, or local environmental laws, regulations, permits, and other elements of the Contractor's Environmental Protection Plan. The Contractor shall, after receipt of such notice, inform the Contracting Officer of the proposed corrective action and take such action when approved by the Contracting Officer. The Contracting Officer may issue

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a stop work order until satisfactory corrective action has been taken. No time extensions shall be granted, or equitable adjustments allowed to the Contractor for any such suspensions. This is in addition to any other actions the Contracting Officer may take under the contract, or in accordance with the Federal Acquisition Regulation or Federal Law.

The Environmental Protection Plan shall include, but is not limited to, the following:

- a) Name(s) of the person(s) within the Contractor's organization who is (are) responsible for ensuring adherence to the Environmental Protection Plan.
- b) Name(s) and qualifications of the person(s) responsible for manifesting hazardous waste to be removed from the site, if applicable.
- c) Description of the Contractor's environmental protection personnel training program.
- d) A Storm Water Pollution Prevention Plan, which identifies the type and location of the erosion and sediment controls to be provided.
- e) Drawings showing locations of proposed temporary excavations or embankments for haul roads, stream crossings, material storage areas, structures, sanitary facilities, and stockpiles of excess or spoil materials, including methods to control runoff and to contain materials on the site.
- f) Traffic Control Plans including measures to reduce erosion of temporary roadbeds by construction traffic, especially during wet weather. Plan shall include measures to minimize the amount of mud transported onto paved public roads by vehicles or runoff.
- g) Work Area Plan showing the proposed activity in each portion of the area and identifying the areas of limited use or nonuse. Plan should include measures for marking the limits of use areas including methods for protection of features to be preserved within authorized work areas.
- h) A Borrowing Material Plan including drawings showing the location of waste and borrow areas.
- i) A Spill Control Plan that includes the procedures, instructions, and reports to be used in the event of an unforeseen spill of a substance regulated by 40 CFR 68, 40 CFR 302, 40 CFR 355, and/or regulated under federal, state, or local laws and regulations. The Spill Control Plan supplements the requirements of EM 385-1-1 and Installation specific measures.
- j) A Solid Waste Disposal Plan identifying methods and locations for solid waste management/disposal including non-hazardous waste streams, soils, and other clearing debris. The plan shall include details for waste stream segregation, management, storage, sampling, handling and schedules for disposal, subcontractors responsible for the transportation and disposal of solid waste, and licenses or permits for solid waste disposal sites that are not a commercial operating facility. Attach evidence of the disposal facility's acceptance of the solid waste to this plan during the construction. Attach a copy of

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each of the Non-hazardous Solid Waste Diversion Reports to the disposal plan. Submit the report on regular intervals (i.e., first working day after the first quarter that non-hazardous solid waste has been disposed and/or diverted and for the previous quarter (e.g., the first working day of January, April, July, and October)). The report shall indicate the total amount of waste generated and total amount of waste diverted in cubic yards or tons along with the percent that was diverted.

- k) A Recycling and Solid Waste Minimization Plan with a list of measures to reduce consumption of energy and natural resources. The plan shall detail the Contractor's actions to comply with and to participate in federal, state, and local government sponsored recycling programs to reduce the volume of solid waste at the source.
- l) An Air Pollution Control Plan detailing provisions to minimize airborne dust, debris, materials, trash, etc., from migrating off site.
- m) A Contaminant Prevention Plan that identifies potentially hazardous substances to be used on the job site; identifies the intended actions to prevent introduction of such materials into the air, water, or ground; and details provisions for compliance with federal, state, and local laws and regulations for storage and handling of these materials. In accordance with EM 385-1-1, a copy of the Safety Data Sheets (SDS) and the maximum quantity of each hazardous material to be on site at any given time shall be included in the Contaminant Prevention Plan. As new hazardous materials are brought on site or removed from the site, the plan shall be updated.
- n) A Wastewater Management Plan that identifies the methods and procedures for management and/or discharge of waste waters which are directly derived from construction activities, such as concrete curing water, clean-up water, decontamination water, hydrostatic test water, and water used in flushing of lines. Include any applicable permits, documentation, and sketches.
- o) A Historical, Archaeological, Cultural, Biological, and Wetland Resources Plan that defines procedures for identifying and protecting the resources known to be on the project site and any resources discovered during construction/demolition. The plan shall identify lines of communication between Contractor personnel and the Contracting Officer.

2.3. Existing Environmental Contamination

Lead based paint (LBP) and/or asbestos containing materials (ACM) may be present at this location in or on structures scheduled for demolition or renovation. The Contractor shall perform a survey for LBP and ACM prior to any demolition or renovation activities. Upon completion of the survey, the Contractor shall share the results with the Contracting Officer, who will determine whether to exercise the associated option CLINs.

2.4. GENERAL DESCRIPTION

Environmental protection is the prevention/control of pollution and habitat disruption that may occur to the environment during construction/demolition. The control of environmental pollution and damage requires consideration of

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land, water, and air; biological and cultural resources; and includes management of visual aesthetics; noise; solid, chemical, gaseous, and liquid waste; radiant energy and radioactive material as well as other pollutants.

2.5. DESIGN CRITERIA AND REFERENCE PUBLICATIONS

UNIFIED FACILITIES CRITERIA (UFC)

UFC 3-460-01 Petroleum Fuel Facilities, Design
(Jul 2019, Chg. 3 - June 2023)

CODE OF FEDERAL REGULATIONS (CFR)

29 CFR 1910 Occupational Safety and Health Standards

29 CFR 1910.120 Hazardous Waste Operations and Emergency Response

29 CFR 1910.1000 Air Contaminants

29 CFR 1926 Safety and Health Regulations for Construction

29 CFR 1926.65 Hazardous Waste Operations and Emergency Response

40 CFR 112 Spill Prevention Control and Countermeasure Plans

40 CFR 261 Identification and Listing of Hazardous Waste

49 CFR 171 General Information, Regulations, and Definitions

49 CFR 172 Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 Safety and Health Requirements Manual (2024)

EP 1110-1-16 Handbook for the Preparation of Storm Water Pollution Prevention Plans for Construction Activities

AMERICAN PETROLEUM INSTITUTE (API)

API RP 2219 (2005) Safe Operation of Vacuum Trucks in Petroleum Service

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 30 (2021) Flammable and Combustible Liquids Code

2.6. COMPLIANCE

a. Comply with and obtain all applicable federal, state, and local regulations and permits required for this project and review federal, state and installation policies and procedures.

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b. Project Design shall be completed pursuant to 40 CFR Parts 112 and/or 280 (as applicable), to the greatest extent possible and in accordance with 29 CFR 1910, 29 CFR 1926, and USACE Safety Regulation EM-385-1-1.

2.7. SAFETY AND HEALTH GENERAL DESIGN

2.7.1. Accident Prevention Plan/Site Safety and Health Plan (APP/SHHP)

Contractor shall be required to prepare a written site-specific Accident Prevention Plan (APP) that addresses all occupational safety and health hazards associated with the project. Should initial sampling and laboratory analysis show contamination over regional screening levels a Site Safety and Health Plan (SSHP) for contaminant-related hazards shall be added to the APP as an appendix. Contractor shall cover each SSHP element in Appendix A of EM 385-1-1 as applicable. The SSHP will require government Contracting Officer acceptance before any work begins.

The Accident Prevention Plan (APP), required as part of the base task order, shall include an Activity Hazard Analysis (AHA) to provide for the safety of workers performing operations which involve exposure to PFAS & petroleum-contaminated media (if applicable). The Contractor shall develop the AHA in accordance with appropriate federal, state, and local regulations.

2.7.2. Site Safety and Health Officer

The Site Safety and Health Officer (SSHO) required in the 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS specification must meet the qualifications listed in this section and shall be in accordance with EM 385-1-1 Section 01.A.17 and Section 33.C.02.

2.7.3. Hot Work and Potential Sources of Ignition

Hot work and potential sources of ignition are of particular concern on this project and will require coordination with the installation fuels safety personnel through the Contracting Officer. Before heat is applied to a drum, container, or hollow structure, a vent or opening shall be provided for the release of any built-up pressure generated. SSHO and designee shall conduct air monitoring for flammable vapors in the work area to ensure the safety of workers, including, at a minimum, the measurement of explosive vapor levels in all work areas using a Combustible Gas Indicator (CGI). Eliminate all ignition sources and prevent the discharge of static electricity during venting of flammable vapors. Use explosion-proof, non-sparking equipment and tools when cutting piping. Clean up all fuel product discharges or spills promptly, and handle and dispose of spill materials, rags, and any petroleum-soaked wastes properly.

2.8. ENVIRONMENTAL DESIGN REQUIREMENTS

The Environmental Design shall include, but is not limited to, the following:

- a. Sampling is required if contamination is suspected during excavation operations for the project.
- b. Design shall include a caveat that if material not indicated that may be hazardous to human health upon disturbance during construction operations is encountered, the SSHO shall stop work immediately and contact the Contracting Officer.

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c. Contractor shall be responsible for verification of all existing conditions prior to demolition. Upon receipt from the Department of Defense (DoD) Environmental Laboratory Accreditation Program (ELAP) approved laboratory, all sampling and lab results will be coordinated through, and documentation will be provided to the Contracting Officer and the Installation Environmental Office via the Contracting Officer.

d. Anticipated sample locations, testing methodology and data interpretation scenarios shall be addressed in the Project specifications and drawings generated, as applicable. All methods and procedures shall follow all federal, state, and local regulations.

e. Laboratory samples collected shall be analyzed by a DoD ELAP certified laboratory. Chemical parameters, analytical methods/procedures, timing constraints, and data interpretation scenarios are presented in a Quality Assurance Project Plan.

f. Any waste descriptions shall be reviewed by an installation Environmental Management Office Point of Contact prior to manifesting. All environmental notifications, permits, permit application packages, fees, approvals to construct, certifications, laboratory analysis, weight bills, and disposal manifest(s) shall be provided to the Contracting Officer and the Installation Environmental Office via the Contracting Officer.

g. Identify locations of monitoring well(s) in the construction/demolition area and vicinity affected by construction/demolition activities. Avoid disturbing the integrity of existing monitoring well(s) during this project. Indicate in specifications and drawings the locations of well(s) that are to be avoided. Address proposed temporary excavations or embankments for haul roads, Contaminant of Concern (COC) locations or plumes, anticipated excavation footprint, and lay down areas for stockpiles of excess or spoil materials.

h. Include the following Unified Facilities Guide Specifications (UFGS), as applicable:

01 35 26	GOVERNMENTAL SAFETY REQUIREMENTS
01 57 19	TEMPORARY ENVIRONMENTAL CONTROLS
01 74 19	CONSTRUCTION AND DEMOLITION WASTE MANAGEMENT
02 41 00	DEMOLITION AND DECONSTRUCTION
02 61 13	EXCAVATION AND HANDLING OF CONTAMINATED MATERIAL
02 81 00	TRANSPORTATION AND DISPOSAL OF HAZARDOUS MATERIALS
02 82 00	ASBESTOS REMEDIATION
02 83 00	LEAD REMEDIATION

i. Contractor shall be responsible for any sampling and final documentation required, in a timely manner. Schedules established shall ensure that the Installation via the Contracting Officer has ample time to review all documentation and submit according to applicable regulatory schedules.

j. Contractor to identify critical tasks, establish communication lines, and a notification timeline for critical portions of the project via the Contracting Officer. The timeline will ensure the availability of regulatory agencies and installation personnel to be on site to observe the critical tasks operations, as required.

2.9. WASTE-STREAM REQUIREMENTS

a. Identify requirements for waste identification, excavation and monitoring, segregation, sampling, storing/handling/disposal methods for hazardous and non-hazardous materials as applicable. Identify requirements for any hazardous waste streams such as contaminated soils or water that requires licenses or permits for transportation and disposal. Sampling requirements for individual waste disposal facilities shall be the responsibility of the Contractor. Evidence of the disposal facility's acceptance of the waste shall be submitted to the Contracting Officer and installation Environmental Management Office via the Contracting Officer at regular intervals. Reports shall indicate the total amount of waste generated and total amount of waste diverted in cubic yards or tons along with the percent that was diverted.

b. Contractor shall notify the Contracting Officer if any material to be disposed of is found to contain hazardous, toxic, biological, or radiological substances. Rubbish and debris will be removed from Government property daily to avoid accumulation at the project site.

c. Identify the methods and procedures for management and/or discharge of waste waters which are directly derived from construction/demolition activities, such as concrete curing water, clean-up water, disinfection water, hydrostatic test water, and water used in flushing of lines. Include any applicable permits, documentation, and sketches.

2.9.1. Lead Based Paint and Paint Containing Chromium (CLIN 0003)

Lead-Based Paint (LBP) and paint containing chromium may be present at this location. If LBP is present, the optional CLIN will then be implemented to address proper removal and handling according to federal, state and local laws and regulations before demolition activities commence.

2.9.2. Asbestos Containing Material (CLIN 0002)

Asbestos Containing Material (ACM) may be present during this project. If ACM is present, the optional CLIN will then be implemented and then it must be removed and handled according to federal, state and local laws and regulations before demolition activities commence.

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3. ARCHITECTURAL SCOPE OF WORK

3.1. GENERAL

A detailed description of the scope of work for the facility identified and items of work are included herein. The Design/Build Contractor will be required to work directly with the base Design Working Group to complete the concept plans provided as concepts for this project.

3.1.1. Site Conditions

Concept plans and information provided herein have been developed from limited on site investigation of the existing site and buildings. The government has no existing building plans for the current conditions and interior configuration of the building to be renovated. The plans included are conceptual in nature and on-site investigation of the existing site and facilities will need to be done by the Design/Build Contractor before beginning their required design deliverables per the contract. All critical dimensions and clearance shall be coordinated to ensure accurate placement of interior spaces and room layouts required.

3.1.2. Wood Construction

Wood construction shall NOT be permitted for any major portion of this design or construction of the interior spaces or the addition.

3.1.3. Framing

Light gage steel studs/framing will be used as indicated in the drawings for the interior non load bearing wall construction. The existing structural systems and exterior envelope are to remain intact, except for new windows, doors, or other openings needed, or as noted otherwise herein. If wood is used for flashing, blocking, nailing strips, roof supports, or other minor items the wood should be a treated type wood.

3.1.4. Site Location

The site is located: At Marine Corps Air Station (MCAS) Miramar, CA.

3.2. DESIGN CRITERIA

Design/Build Contractor shall provide the following design submittals as described below. Include all requirements as noted in the latest version of the Omaha District Design Guide (ODDG). For minimum drawing, narrative and other criteria refer to the Architectural Design Chapter.

Preliminary Design - 65%, which shall include the following:

- General: The purpose of the Preliminary Design phase is to validate that the design is being developed in accordance with the approved design criteria and is meeting the customer's expectations.
- Drawings
 - Shall be developed to a 65% design.
 - All floor and ceiling plans shall be nearly complete. Floor plans shall show complete functional relationships of spaces, required adjacencies, circulation, required exiting, and required number of plumbing

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fixtures. Plans shall have established all major and secondary dimensions, identify wall types and section cut locations, door and window types, interior elevations directions, ceiling types and lighting established, and completed with wall poche. Include interior furniture footprint plans of appropriate areas where necessary to determine the functionality of the spaces.

- o All wall sections at all conditions shall be fully coordinated and completed.
- o All schedules (Door, Window, Louver, Room Finish) shall be complete, except for door hardware sets information.
- o All wall types shall be fully coordinated and completed.
- o Detailing of major building elements shall be started.
- o Roof plans shall be nearly complete.
- o Exterior elevations shall be nearly complete.
- o Interior elevations shall be nearly complete.
- o Life safety plan shall be complete.
- Specifications: At 65% level.
- Design Analysis: Prepared and to include information listed in the DESIGN ANALYSIS Paragraph of the Architectural Design Chapter and the general criteria indicated in Chapter 17 of the ODDG.

Final Design - 100%, which shall include the following:

- General: The purpose of the Final Design phase is to review for final completion of all project documents and to validate that the design is being completed in accordance with the approved design criteria and customer's requirements. And incorporate into the design all previously approved review comments.
- Drawings
 - o The drawings shall be complete, thoroughly checked, and fully coordinated with the specifications and all other construction documents. Previous comments and applicable criteria changes shall have been incorporated into the design. The contract drawings shall fully describe the type and the scope of work required. All detailing shall be complete.
- Specifications: Shall be completely edited and fully coordinated with the drawings. Specifications shall be edited format showing deletions and insertions for review.
- Design Analysis: Shall include the required information presented in previous submittal, updated to reflect the final design development and corrected to reflect changes in content made in response to review comments and/or scope changes.

Issued for Construction (IFC) Documents

- Propose: The Issued for Construction (IFC) phase is to review for 100% completion of all project documents. Drawings and specifications shall be completely prepared and ready for construction. The final form of the contract documents shall make it possible for construction of the facilities under this contract. The following information shall be submitted for review and approval.
- Drawings: Shall be complete. All previous approved comments shall be incorporated into the drawing set.
- Specifications: Shall be complete and fully coordinated with the drawings. Specifications shall be in a "clean" format without deleted items showing.

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- Design Analysis: Shall be complete. All previous design criteria, calculations, and other design development information shall be included in the Design Analysis. The Design Analysis shall serve as a historical record for documentation of design development, design rational and decisions made, and customer requirements and directions.

The Codes, Standards, and Regulations listed below shall be used in the construction of this project. All applicable local, state, and federal standards and codes shall also be followed. The publications shall be the most recent editions. Standards other than those mentioned may be accepted provided they meet the minimum requirements of the following and the Contractor shall submit proof of equivalency to the Contracting Officer for approval.

International Code Council (ICC):

IBC - International Building Code, as modified by UFC 1-200-01.

National Fire Protection Association (NFPA)

NFPA 101 - Life Safety Code

NFPA 80 - Standard for Fire Doors and Fire Windows

NFPA 10 - Standards for Portable Fire Extinguishers

NFPA 24 - Standard for the Installation of Private Fire Service Water Mains

NFPA 13 - Standard for the Installation of Sprinkler Systems

NFPA 72 - National Fire Alarm Signaling Code

Code of Federal Regulations (CFR)

CFR 1910.132; Personal Protective Equipment (PPE)

CFR 1910.157; Portable Fire Extinguishers

CFR 1910, Subpart D; Walking-Working Surfaces

CFR 1910, Subpart E; Means of Egress

U.S. Department of Defense (DOD): Unified Facilities Criteria (UFC)

UFC 1-200-01; DOD Building Code (General Building Requirements), with Change 2, dated 12 June 2023

UFC 1-200-02 High Performance and Sustainable Building Requirements, with Change 2, dated 1 June 2022

UFC 3-101-01; Architecture, with Change 3, dated 2 Oct 2023

UFC 3-110-03; Roofing, with Change 5, dated 12 June 2020

UFC 3-120-10; Interior Design, with Change 2, dated 15 June 2021

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UFC 3-120-01; Design: Sign Standards, with Change 4, dated 1 Feb 2023

UFC 3-400-02; Design: Engineering Weather Data, dated 20 September 2018

UFC 3-600-01; Fire Protection Engineering for Facilities, with Change 6, dated 6 May 2021

UFC 4-010-01; DOD Minimum antiterrorism standards for buildings, with Change 2, dated 30 July 2022

UFC 4-310-03; DOD Fuels Laboratory Standard, dated 12 Aug 2020

UFC 4-021-01; Design and O&M: Mass Notification Systems, with Change 1, dated 1 Jan 2010

ASHRAE Standards and Handbook

SMACNA Standards and Handbook

USACE Unified Facility Guide Specifications (UFGS)

21 13 13.00-10; Wet Pipe Sprinkler System

28 31 64-10; Fire Detection and Alarm System

28 31 76; Interior Fire Alarm and Mass Notification System

USACE: Engineering and Construction Bulletin (EBC)

2014 Leadership in Energy and Environmental Design (LEED), Version 4 (V4) - Applicability: Information and Guidance

2013-5 Continuity of the High-Performance Energy and Sustainability Policy - Applicability: Directive

2012-16 Building Air Tightness and Air Barrier Continuity Requirements

2008-1 Sustainability Design & Development (SSD) and Subsequent ECB's Related to Sustainability and Energy Reduction

3.3. DEMOLITION

Demolition scope of facility is described below. Refer to the civil scope of work to provide information on site demolition. Disposal of all removed materials shall be outside the limits of Government-controlled land in accordance with Federal, State, and local regulations. The Contractor shall notify the Contracting Officer's Representative if any materials that are to be disposed of are found to contain hazardous, toxic, biological, or radiological substances.

The Design/Build Contractor will hire as part of their contract an environmental engineer to determine the presence of hazardous material in these facilities and the proper removal and disposal procedures to be followed by the Design/Build Contract during construction. If lead base paint is present it is currently encapsulated with new paint. Do not scrap or sand existing painted surfaces until it is determined by the environmental engineer if lead-based paint is present and if so how to safely remove it. It

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was observed that the existing Laboratory has Vinyl Asbestos Tile and likely the mastic contains ACM in that flooring. The Owner representative was unable to provide additional information on hazardous materials present on site so the Design/Build Contractor shall have an Asbestos and Lead-Based Paint Inspection Report completed for each of the facilities that will require removal of hazardous materials if present.

3.3.1. DEMOLITION TO CONSTRUCT/RENOVATE FUEL LABORATORY FACILITY 9647 (CLIN 0001)

FUEL LABORATORY FACILITY: This building will be completely renovated. The existing building wall structure is concrete clad with an adobe block finish, and the roof structure is precast concrete which shall remain. Remove the existing adobe block finish to be replaced with CMU block to match the adjacent Fuel Dispatch Facility. Any areas of existing wall or roof deck structure where the reinforcing is exposed shall be repaired. All exterior windows, doors, and louvers shall be replaced. Roofing is a BUR with granular cover. Existing roof to be replaced by completely removing the existing roofing and insulation down to the existing roof deck along with all associated metal flashings. The existing HVAC system was recently replaced and will need to be protected during all roof repair efforts. Existing interior spaces completely gutted and all new finishes provided.

3.4. LIFE SAFETY / FIRE PROTECTION / HANDICAPPED ACCESSIBILITY

These facilities shall be designed in accordance with recognized industry standards for life safety and building egress. Fire extinguishers cabinets and fire extinguisher brackets including fire extinguishers shall be provided as required per NFPA 10. The facility to be renovated is classified as a Type C++ lab facility per UFC 4-310-03 and anticipates to accommodate abled-bodied personnel only, therefore, the facility does not need to be designed in accordance with ABA requirements.

3.5. BUILDING REQUIREMENTS:

3.5.1. Building: Construct/Renovate Fuel Laboratory Facility 9647 (CLIN 0001)

Background: The Fuels Laboratory Facility 9647 is located on the Flightline. This facility includes a restroom, office, and lab complete with fume hood, testing equipment and storage cabinets. The facilities building materials, roof and systems are degraded, aged, and are in need of immediate replacement/renovation. This project will provide a complete gut and renovation of the existing fuel dispatch facility building to meet the mission requirements for the Base.

- Fuel Lab is located on the Flightline adjacent to the Fuel Dispatch.
- The exterior of the building is clad with an adobe block construction, and the existing roof structure is concrete. Exterior adobe block to be removed and replaced with new CMU with integral color, and water repellent to match the adjacent Fuel Dispatch Facility.
- Roofing is a BUR with granular ballast. Existing roof to be replaced by completely removing the existing roofing and insulation down to the existing roof deck along with all associated metal flashings. Replace existing roof

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membrane and insulation with new rigid roof insulation and new single membrane roofing system per the base design standards, including all associated roof flashings.

- Exterior doors and windows to be replaced. This will include infilling of some existing doors, windows and exterior openings and installation of new doors, windows, and openings in new locations to accommodate a new functional layout of interior spaces.
- The floor slab will need to have containment provided by recessing the floor slab in Laboratory 101 from the exterior of the building and from the Laboratory Office 100. This containment could also be provided by raising the Laboratory Office 100 floor slab 6-inch and providing a raised curb at the exterior door to the building with an exterior ramp. UFC 4-310-03 requires a 6-inch recessed floor slab and floor drain be provided in the Fuels Testing Lab 101 and this requirement must be met.
- The interior finishes have seen their life expectancy and will be replaced.
- The existing Fuel Laboratory Facility is approximately 18' wide x 25' long x 12' high and is 450 SF.

Concept plans have been provided per Attachment E - RFP Drawings which are to be used by the Design-Build Contractor during the design phase of the project with the user groups as a starting point for discussion with the base when looking at the possibilities for the new building layouts. These concept plans shall not be considered final but are used to show users ideas of what these spaces should include at a minimum.

The building has no IT room or vestibules and given space is extremely tight additions might need to be considered for these types of spaces if required to be added to the existing building.

3.5.1.1. Laboratory Office:

Approximately 140 square feet.

- Provide the general location of the office furniture and equipment location in design documents.
- Floor finish shall consist of porcelain tile.
- Walls consisting of 4-inch steel studs covered with 5/8 inch gypsum wallboard at interior wall locations, at existing perimeter wall location new walls consisting of 2-inch insulated Z-furring with 2" continuous polyisocyanurate rigid board insulation cover with 5/8 inch gypsum wallboard.
- New walls shall be primed and painted with two finish coats of semi-gloss paint.
- Ceiling shall consist of a GWB ceiling system.

3.5.1.2. Laboratory:

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Approximately 240 square feet.

- Provide the general location of the office furniture and equipment location in design documents.
- Floor finish shall consist of fuel resistant epoxy resinous flooring and base. Note floor to be recessed 6-inches per UFC 4-310-03 and provided with a pluggable floor drain in Fuels Testing Lab 101.
- Walls consisting of 4-inch steel studs covered with 5/8-inch gypsum wallboard at interior wall locations, at existing perimeter wall location new walls consisting of 2-inch insulated Z-furring with 2" continuous polyisocyanurate rigid board insulation cover with 5/8 inch gypsum wallboard. Wall construction to be full height, minimum 1-hour fire rated and UL fire-resistant.
- New walls shall be primed and painted with two finish coats of semi-gloss paint.
- Ceiling shall consist of a 24" x 24" x 5/8" suspended acoustical tile ceiling system.
- Comply with latest edition of the DOD Fuels Laboratory Standard - UFC 4-310-03 for the Basic Lab requirement and the Type C++ Lab requirements, and furnishings and equipment to include but not limited to the follow:
 - o Emergency Shower and Eye Wash
 - o Double Bowl Wash Sink
 - o Dish Washer (and hot water temperatures)
 - o Fire Extinguisher & Cabinet
 - o Balance Table
 - o Desk
 - o Fume Hood
 - o Hanging Steel Cabinets with Adjustable Shelves and Glazed Doors
 - o Steel Base Cabinets with Adjustable Shelves and Drawers
 - o Epoxy Resin Countertops and Backsplash
 - o Soap and Paper Towel Dispensers

3.5.1.3. Toilet /Locker Rooms:

This facility will use the adjacent Fuel Dispatch Facility for restroom.

3.5.1.4. Mechanical Room:

- The existing building does not have a mechanical room since all equipment is located on roof or at grade around the perimeter of the existing building.

3.5.1.5. Communications Room:

- Communication Room is not provided in the existing building. Provide as required to accommodate all required equipment.

3.6. CONSTRUCTION AND FINISHES

3.6.1. Exterior: Finishes shall follow base standards for this

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installation and selected by the contracting officer.

3.6.1.1. Fuel Laboratory Facility 9647 (CLIN 0001): This is a renovation of the existing masonry building.

The existing exterior adobe block to be replaced with new CMU with integral color to complement the adjacent Fuel Dispatch Facility.

Doors will be painted to match the window color.

Windows will be fixed, thermally broken aluminum windows with prefinished color to match the base aesthetics standards.

Roof edge flashing to match the window and door color.

All exiting painted metal surfaces attached to the building or adjacent bollards to remain will be cleaned, prepped, and repainted to match the existing color and or the new base standard color for each item.

The floor of the laboratory will need to have a pluggable floor drain and containment provided either by recessing the floor or having a raised curb provided at the doors to the building with an accessible exterior ramp and raising the laboratory office area.

Exterior Signage: Wall mounted exterior signage specified to conform to the minimum requirements and allowable options within the guide specifications SECTION 10 14 00.10 EXTERIOR SIGNAGE. Adjacent to the main entrance to the Laboratory provide a 16" x 16" sign per base standards with 3" high Building Numbers and 2" high letter for Building Name. Colors per base standards. At the entrance door to the Laboratory Office provide a sign 12" wide x 6" high with 3" high Room Number and 1/2" high Room Name letters.

3.6.2. Insulation for Exterior Walls and Roofs:

Insulation for exterior walls shall be located on the interior side and shall provide a total R-value of 12 minimum for the insulation and a total U-value of 0.067 for the exterior wall construction of as follows:

R-Value Calculation for Existing and New Building for Masonry Walls:

Outside air film	0.17
8" CMU (or Conc.)	1.50
Insulation	12.00
5/8" GWB	0.56
<u>Inside air film</u>	<u>0.68</u>
Total R =	14.91
U-Value = 1/14.91 =	0.067

Insulation for roofs shall be located entirely above the roof deck and shall provide a total R-value of (25) minimum for the insulation and a total U-value of 0.038 for the exterior wall construction of as follows:

R-Value Calculation for Existing and New Building for Masonry Walls:

Outside air film	0.17
Roof Insulation	25.00
<u>Inside air film</u>	<u>0.68</u>

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Total R = 25.85
U-Value = $1/25.85 = 0.038$

3.6.3. Interior walls:

Gypsum Wallboard and Metal Stud Furring System: Specified to conform to the minimum requirements and allowable options within the guide specifications SECTION 09 29 00 GYPSUM BOARD. Painted gypsum board walls exposed to view shall have a medium knock down texture.

3.6.4. Ceilings:

Suspended Acoustical Tile Ceiling Systems: Suspended acoustical tile ceiling systems specified to conform to the minimum requirements and allowable options within the guide specifications SECTION 09 51 00 ACOUSTICAL CEILINGS. Exposed suspended ceiling grid systems to be 24" x 24" grid module. All suspended ceiling systems must be seismically braced. Locations include offices, corridors, conference rooms, break rooms, training, and storage rooms.

Suspended Gypsum Board System: Suspended gypsum board ceiling systems specified to conform to the minimum requirements and allowable options within the guide specifications SECTION 09 29 00 GYPSUM BOARD. Suspended painted gypsum board ceiling systems provided in restrooms, utility rooms, janitors closets, and vestibules.

Exposed Structure and Equipment: Exposed structure and equipment painted to conform to the minimum requirements and allowable options within the guide specifications SECTION 09 90 00 PAINTING AND COATINGS. Exposed to view structural elements, ceiling hung equipment and exposed piping and ductwork in the electrical and mechanical rooms to be painted.

3.6.5. Doors:

Exterior Personnel Doors: Hollow metal doors specified to conform to the minimum requirements and allowable options within the guide specifications SECTION 08 11 13 STEEL DOORS AND FRAME. All hollow metal door and frame assemblies constructed as required by ANSI A250.8 to meet or exceed an extra heavy duty, grade III seamless-hollow steel construction, and constructed with flush end enclosures at the top and flush enclosures or recessed channels at the bottom. Hollow metal interior doors, except at utility or storage rooms provided with vision panels. Hollow metal doors to be insulated with a foamed-in-place polyurethane core and frames galvanized and field or factory painted.

Doors at Main and Secondary Entrances: Hollow metal doors same as the Exterior Personnel Doors and paint finish to match the aluminum windows prefinished color per base standards. All exterior door assemblies to be weather-stripped and provided with metal thresholds. All personnel door openings are a minimum of 3'-0" wide x 7'-0" high. All entrance doors to be provided with half glass or full glass.

Door Hardware: Door hardware was specified to conform to the minimum requirements and allowable options within the guide specifications SECTION 08 71 00 DOOR HARDWARE.

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Locksets, Panic Devices and Latch Sets: Comply with BHMA Series 4000, Grade I. All locksets, panic devices and padlocks comply with Base standards Best Locksets and are provided with 7 pin removable cores. A minimum of two keys for each lockset were provided. Keys were provided to the Contracting Officer organized in a key control system conforming to ANSI A156.5. The key control storage box was installed where directed by the Contracting Officer or his authorized representative. The successful contractor provided construction cores during construction.

Hardware Finishes and Materials: Finishes and materials are per ANSI A156.18 and the finishes are Satin Chrome.

Door Accessories: All exterior doors fully weather-stripped and include metal kick plates where appropriate and heavy-duty metal thresholds. All exterior doors have non-removable hinge pins. Door bumpers provided for all doors without closers. Door closers, where required were designed in accordance with ANSI A156.4 and NFPA 80 Fire Doors and Windows.

Master Keys: Master keying and individual room keying requirements are in accordance with the instructions provided by the facility users at the required pre-delivery conference.

3.6.6. Windows:

Exterior Windows: Exterior windows with manufacturer's standard thermal break type, double-glazed product whose thermal performance complies with the U.S. Department of Energy, Energy Star Window Program for the Southern Climate Zone. Windows meet the minimum construction requirements for structural performance, 6mm minimum thickness of interior room side glass pane to meet DoD ATFP standard for force protection, air infiltration, water penetration, minimum STC, and condensation index specified conforms to the minimum requirements and allowable options within the guide specifications SECTION 08 51 13 ALUMINUM WINDOWS.

Window Finish: Exposed surfaces of aluminum windows finished with anodic coating conforming to AA DAF-45: Architectural Class I AA-M10-C22-A44, color per base standards, anodic coating 0.7 mil or thicker.

Glazing Units: Glazing units for all exterior windows double glazed fixed, sealed, an outboard lite of ¼-inch thick annealed glass with a minimum ½-inch air space, and a clear inboard lite of ¼" thick annealed laminated glass. The outboard lite to be safety or tempered glazing where required by safety or Life Safety requirements.

Interior Windows, Transoms, and Vision Lites: Window frames aluminum or painted steel. Glazing for interior windows to be a minimum of ¼-inch thick clear or obscured annealed laminated glass and sealed. Glazing to be safety or fire rated where required by safety of NFPA 80 requirements. Glazing to be specified to conform to the minimum requirements and allowable options within the guide specifications SECTION 08 81 00 GLAZING.

3.6.7. Exterior finishes:

Concrete: Architectural precast concrete limited to precast sills. Concrete work limited to the structural and civil portions of the project.

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Masonry: Exterior CMU consisting of integrally colored units in finish, pattern and texture to comply with base standards. Units to be standard 4" x 8" x 16" modular veneer and or 8" x 8" x 16" standard CMU as required. Units and mortar include appropriate admixture to control efflorescence. Interior CMU as required to be standard modular sized units with a smooth face.

Masonry Water Repellant: Concrete masonry units specified with manufacturer's integral water repellant. Additionally, all exterior masonry surfaces are specified to be sealed against water intrusion using an impregnating clear sealer.

Masonry Coursing: Building dimensions and locations of openings designed to course with standard modular CMU, thereby eliminating the need to cut or masonry units as much as possible. The mortar color matches the masonry unit integral color. The mortar joints to be flush cut or raked on split faced unit and concave for other types of masonry units.

Miscellaneous Ferrous Metal: Ferrous metals were factory primed to the greatest extent possible. Dissimilar metals were separated as required to avoid galvanic actions. Ferrous metals exposed to view which do not have a factory finish to be primed according to the requirements of SECTION 09 90 00 - PAINTS AND COATINGS.

Exterior Aluminum: Exterior aluminum exposed to view to be a factory finished anodized aluminum Class I finish per Aluminum Association Designation System. Where multiple metal components were exposed to view such as storefront door and window systems, louvers, flashings or copings, the finish, texture and color were visually correlated and were provided by a single manufacturer to the greatest extent possible.

Roof Membrane: Comply with base standards for a single ply full adhered roof membrane system.

Roof Insulation: Roof insulation to be designed as continuous over the entire roof area and specified to conform to the minimum requirements and allowable options within guide specification SECTION 07 22 00 ROOF AND DECK INSULATION. Minimum R-25 continuous insulation above the deck provided for all roof areas per the 2015 International Energy Conservation Code (IECC).

Roof Drainage: Gutters and downspouts were utilized to control the flow of water off of the metal roofs. Roof flashing was specified to conform to the minimum requirements and allowable options within the guide specifications SECTION 07 60 00 FLASHING AND SHEET METAL.

Roof Penetrations: Roof penetrations to be minimized as far as practicable and sealed or otherwise waterproofed. Where penetrations are uniform such as pipes or tubes, bell cap flashings, fasteners, and pre-manufactured neoprene flashings to be specified as applicable.

3.6.8. Interior finish:

Paint Systems: Paint systems specified to conform to the minimum requirements and allowable options within the guide specifications SECTION 09 90 00 PAINTING AND COATINGS. Paint shall have less than .06% lead by weight. Paint color shall be selected from manufacturer's standard samples.

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Wall Base: Resilient wall base specified to conform to the minimum requirements and allowable options within the guide specifications SECTION 09 65 00 RESILIENT FLOORING. Resilient wall base specified for all office areas utilizing porcelain tile flooring or sealed concrete flooring. Porcelain tile cove base to be specified to conform to the minimum requirements and allowable options within the guide specifications SECTION 09 30 00 CERAMIC TILE. Porcelain tile cove base specified for all areas utilizing ceramic tile flooring in vestibules, classroom, break room, restrooms, and locker rooms. Wall base is a minimum of 4-inches high.

Corner Guards: Vinyl clad edge (corner) guards specified for protection of finishes at locations subject to impacts. Top of edge guard's minimum of 4'0" above finished floor. Bottom of guards are to be at top of base.

Interior Signage: Interior Signage specified to conform to ABA and to the minimum requirements and allowable options within the guide specification SECTION 10 14 00.20 INTERIOR SIGNAGE. Provide signage for all rooms or spaces and refer to UFC 4-310-03 DoD Fuels Laboratory Standards for specific facility signage requirements.

3.6.8.1. Flooring:

Porcelain Tile: Porcelain tile and base specified to conform to the minimum requirements and allowable options within the guide specifications SECTION 09 30 00 CERAMIC TILE. Tiles shall be through body color, not surface applied color or glazed.

Fuel Resistive Resinous Flooring & Base: Epoxy fuel resistive resinous flooring and base to conform to the requirements and allowable options within the guide specification SECTION 09 67 23.16 FUEL RESISTIVE RESINOUS FLOORING, 5-COAT SYSTEM.

4. STRUCTURAL

4.1. GENERAL

The construction and renovation operations must be completed without exceeding the allowable stress for the materials of construction in the structural members and connections. Changes to existing gravity and lateral systems are not anticipated. However, should modifications or reinforcements become necessary, the requirements of this section shall govern all new structural elements and any affected components within the load path.

Existing drawings of Facility 9647 are provided in Attachment G.

4.2. DESIGN ANALYSIS

Design must be performed and signed by a registered professional engineer. All required documents, including drawings, specifications, and basis of design, must be prepared in accordance with Section 01 33 00 Submittal Procedures. Specific submittal requirements in these sections supplement the requirements of Section 01 33 00.

The basis of design must include statements on the following:

- a. List a summary of the criteria upon which the structural design is based. Including a statement of all loads: dead, live, wind, snow, earthquake, and any other significant load.
- b. List all assumptions required for the structural design where the design criteria are undefined, unclear, conflicting or unknown. State the basis of the assumption made.
- c. Describe the structural system. Include a discussion of both the gravity and lateral force resisting systems. Clearly describe the gravity and lateral load paths providing all pertinent information such as, capacity, size, dimensions, materials, and design strengths.
- d. Provide a narrative summary of the foundation system, including method for determination of the bearing capacity, maximum allowable bearing capacity, and lateral force capacity of the foundation, as well as other soil parameters used in the design. Provide all pertinent information, such as capacity, size, dimensions, and a list of materials with design strengths.
- e. When appropriate, provide a statement of any special considerations that affect the design (e.g., special corrosion resistance requirements, fire-resistive requirements, etc.)
- f. Describe applicable special inspections, testing, and observations required per IBC 2021 Chapter 17.

Provide calculations to support all items and details outlined on the drawings and specifications. They must be legible, orderly, easily understood, and checked by a registered practicing structural engineer.

Also include the following in the calculations:

- a. A cover sheet indicating the project title, location, construction contract number, date, and names of the persons originating and checking the calculations.
- b. Table of contents

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- c. A brief statement describing the structural system, significant design parameters and any restrictions that may affect the design.
- d. Applicable design criteria
- e. Loads: Include all loadings, forces, temperature changes, induced settlements, and other internal and external actions that may affect the design of the structure. The list must include the orientation and location, magnitude and units of measure for each load.
- f. Restrictions: Include all limiting factors such as horizontal and vertical deflections limits, height restrictions, installation or operating tolerances for equipment or building components and any other limits to the structural system.
- g. Materials: Include all materials used and their allowable stress limits, yield strengths, type, grade, class or other applicable material properties.
- h. References: Include all criteria, accepted standards, manuals, codes, texts, papers, or other references used in the analysis and design that are accepted in a public domain. All references must be appropriately identified; abbreviations such as AISC, ASTM, and ACI are acceptable. Document the origin of all customer specific criteria in the calculations.
- i. Sketches with sufficient detail and clarity to communicate design intent. Note assumptions and references to codes, standards, criteria drawings, and computer output.

Design for lateral forces must include design calculations for wind, seismic, and other potential loadings. The construction drawings must depict the design elements meeting both the seismic and wind design requirements.

Story drift must not exceed 1/240 times the story height for design wind and seismic load combinations. Structural members must resist lateral force, as determined by a stress analysis on the building. Wood must not be used for structural members.

Computer generated calculations must identify the program name, source, and version. Input must show, as a minimum, nodes/joints, elements/members, materials/properties, and all loadings, induced settlements/deflections, etc. and a list of load combinations. Results must include an output listing for maximum/minimum stresses/forces, deflections for each element, and the restrictions for each loading case and combination.

4.3. DRAWINGS

Provide structural drawings that sufficiently detail all structural work. The drawings must contain a set of "Structural Notes" in accordance with Chapter 16 of IBC 2021. Provide critical reference information for future building modifications or evaluations. As a minimum, provide Design Criteria, Design Standard References, General Construction requirements, and the following:

- a. Loads: Provide loading information and identify sources for all listed loads.
- b. Foundation Conditions: Fully describe the foundation conditions and list the type of foundation system and method employed to determine allowable soil bearing values. Indicate the minimum allowable bearing capacity for shallow foundations, or the pile

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or pier capacity in both tension and compression for deep foundations. Indicate passive, active and at rest design pressures, the coefficient of friction and the sub-grade modulus. Indicate if a site-specific design response spectrum has been used in the design and give the site class in accordance with the seismic design criteria.

- c. Materials: Clearly define the types, grades, and properties of materials for each structural element and system.
- d. Quality Assurance: Provide a summary of the quality assurance requirements.

4.4. SPECIFICATIONS

Provide edited specifications in accordance with the designer notes when sufficient detail is necessary to fully describe and demonstrate the quality of materials, the installation and performance of equipment, and the quality of workmanship. Specifications must not be edited to remove applicable technical requirements that lesson the quality of the material and services being provided. The Designer of Record (DOR) must be responsible for reviewing submittals. See Attachment C - Sample Submittal Paragraph for additional information.

4.5. REFERENCES

The Codes, Standards, and Regulations listed below must be used in the design and construction of this project. All applicable local, state, and federal standards and codes must also be followed. Whenever two design documents are in conflict the most stringent criteria must govern. All design criteria, references and standards used should be current editions at the time of contract award.

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 117 (2010) Specification for Tolerances for Concrete Construction and Materials

ACI 201.2R (2016) Guide to Durable Concrete

ACI 211.1 (2022) Selecting Proportions for Normal-Density and High Density-Concrete - Guide

ACI 302.1R (2015) Guide for Concrete Floor and Slab Construction

ACI 305R (2020) Guide to Hot Weather Concreting

ACI 306R (2016) Guide to Cold Weather Concreting

ACI 318-19; R 2022 Building Code Requirements for Structural Concrete

ACI 360R (2010) Guide to Design of Slabs-On-Grade

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 325 (2023) Steel Construction Manual-16th Edition

AISC 360 (2022) Specification for Structural Steel Buildings

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AMERICAN IRON AND STEEL INSTITUTE (AISI)

AISI S100 (2016) North American Specification for the Design of Cold-Formed Steel Structural Members

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-22 Minimum Design Loads for Buildings and Other Structures,

ASTM INTERNATIONAL (ASTM)

ASTM A6/A6M Standard Specification for General Requirements for Rolled Structural Steel Bars, Plates, Shapes, and Sheet Piling

ASTM A29/A29M Standard Specification for Steel Bars, Carbon and Alloy, Hot-Wrought General Requirements for

ASTM A36/A36M Standard Specification for Carbon Structural Steel

ASTM A53/A53M Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM A123/A123M Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM A153/A153M Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

ASTM A615/A615M Standard Specification for Deformed and Plain Carbon Steel Bars for Concrete Reinforcement

ASTM A992/A992M Standard Specification for Structural Steel Shapes

ASTM A1085/1085M Standard Specification for Cold-Formed Welded Carbon Steel Hollow Structural Sections (HSS)

ASTM C31/C31M Standard Practice for Making and Curing Concrete Test Specimens in the Field

ASTM C33/C33M Standard Specification for Concrete Aggregates

ASTM C39/C39M Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens

ASTM C452/C452M Standard Test Method for Potential Expansion of Portland-Cement Mortars Exposed to Sulfate

ASTM C989 Standard Specifications for Slag Cement for Use in Concrete and Mortars

ASTM C1012/C1012M Standard Test Method for Length Change of Hydraulic-Cement Mortars Exposed to a Sulfate Solution

ASTM C1260 Potential Alkali Reactivity of Aggregates (Mortar-Bar Method)

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ASTM C1567 Standard Test Method for Potential Alkali-Silica Reactivity
of Combinations of Cementitious Materials and Aggregate (Accelerated
Mortar-Bar Method)

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020) Structural Welding Code – Steel

DEPARTMENT OF THE ARMY

EM 385-1-1 (2014) Safety – Safety and Health Requirements

INTERNATIONAL CODE COUNCIL (ICC)

IBC (2024) International Building Code

IEBC (2024) International Existing Building Code

INTERAGENCY COMMITTEE ON SEISMIC SAFETY IN CONSTRUCTION (ICSSC)

ICSSC (RP 8) Standards of Seismic Safety for Existing Federally Owned
and Leased Buildings

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA 30 (2021) Flammable and Combustible Liquids Code

NFPA 30A (2021) Code for Motor Fuel Dispensing Facilities and Repair
Garages

NATIONAL CONCRETE MASONRY ASSOCIATION (NCMA)

NCMA TEK 10-03 (2003) Control Joints for Concrete Masonry Walls –
Alternative Engineered Method

THE MASONRY SOCIETY (TMS)

TMS 402 (2022) Building Code Requirements for Masonry Structures

TMS 602 (2022) Specification for Masonry Structures

UNIFIED FACILITIES CRITERIA (UFC)

(<https://www.wbdg.org/dod/ufc>)

SERIES 1-200: POLICY

UFC 1-200-01 (2024) DoD Building Code, with Change 4

SERIES 3-200: CIVIL / GEOTECHNICAL / LANDSCAPE ARCHITECTURE

UFC 3-220-01 (2025) Geotechnical Engineering, with change 3

SERIES 3-300 – STRUCTURAL AND SEISMIC DESIGN

UFC 3-301-01 (2026) Structural Engineering, with Change 6

SERIES 3-400: MECHANICAL

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UFC 3-460-01 (2023) Design: Petroleum Fuel Facilities, with
Change 3

SERIES 4: MULTI-DISCIPLINARY AND FACILITY-SPECIFIC DESIGN

UFC 4-010-01 (2024) DoD Minimum Antiterrorism Standards for
Buildings, with Change 3

UFC 4-021-01 (2010) Design and O&M: Mass Notification Systems with
Change 1

UNIFIED FACILITIES GUIDE SPECIFICATIONS (UFGS)

(Unedited: <https://www.wbdg.org/dod/ufgs>)

DIVISION 03 - CONCRETE

03 30 00 Cast-In-Place Concrete

04 20 00 Unit Masonry

DIVISION 05 - METALS

05 05 23.16 Structural Welding

05 12 00 Structural Steel

05 40 00 Cold-Formed Metal Framing

05 50 14 Structural Metal Fabrications

05 51 00 Metal Stairs

05 51 33 Metal Ladders

05 52 00 Metal Railings

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

07 92 00 Joint Sealants

Division 21 - FIRE PROTECTION

21 13 13.00-10 Wet Pipe Sprinklers for Fire Protection

Division 28 - ELECTRONIC SAFETY AND SECURITY

28 31 74.00-20 INTERIOR FIRE Detection and ALARM

28 31 76 INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.28 Duty to Have Fall Protection and Falling Object
Protection

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29 CFR 1910.29 Fall Protection Systems and Falling Object Protection -
Criteria and Practices

4.6. DESIGN CRITERIA

4.6.1. Risk Category

The listed facilities must be classified as follows per Table 2-2 in UFC 3-301-01:

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4.6.2. Dead and Live Loads

Dead and live loads used for design must be in accordance with ASCE 7, UFC 3-301-01, and as required by OSHA. The minimum roof live load must be 20 psf.

4.6.3. Snow Loads

Snow loads must be calculated in accordance with UFC 3-301-01 and ASCE 7. The minimum ground snow load must be 4 PSF.

4.6.4. Wind Loads

Wind loads must be calculated in accordance with ASCE 7 using a "3-second gust" wind speed of 96 mph and Exposure Category "C" for Risk Category II structures in accordance with UFC 3-301-01.

4.6.5. Environmental Severity Classification (ESC) Parameters

UFC 1-200-01 requires that ESC corrosion categories be determined for DoD projects. ESC corrosion categories for each installation can be found in Table A-1 of UFC 1-200-01. The ESC Corrosivity Category Estimation Tool (ICCET) can be found within the Corrosion Toolbox on the WBDG at <https://www.wbdg.org/tools/corrdefense/map.html>. Per UFC 3-301-01, sites that are classified as C3-C5 must have additional concrete cover over reinforcing steel provided in accordance with ACI 357.3R Table 5.5.4. Additional corrosion prevention measures are also required for exposed steel and metal works; considerations for steel structures are provided in section A-5.4 of UFC 3-301-01.

4.6.6. Seismic

Seismic loads must be calculated in accordance with ASCE 7. Spectral response acceleration parameters for MCAS Miramar must be $S_s = 1.060$ g and $S_1 = 0.330$ g with Site Class "D" unless Site Class "E" or "F" soils are present at the site. The geotechnical engineer must be permitted to modify the site classification based on the soil properties found from site exploration.

4.7. FORCE PROTECTION

Force protection must comply with the requirements of UFC 4-010-01 with the appropriate exemptions being taken into account.

4.8. CONCRETE

Concrete must be in accordance with provisions of the American Concrete

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Institute, Building Code Requirements for Structural Concrete, ACI 318.

A minimum 28-day compressive strength (f'_c) of 4,500 psi, unless otherwise stated in ACI 318, must be used for design and construction of all structural concrete. Acceptance testing for concrete must be based on four or six-inch diameter cylinders fabricated and tested according to requirements in ASTM C31 and ASTM C39. Cylinder size for acceptance testing must match the cylinder size used in the concrete mix design. Mix design proportioning must be in accordance with ACI 211.1. Concrete must have a water-cement ratio not to exceed 0.45.

Concrete placement during hot weather must comply with recommendations of ACI 305R. Concrete placement during cold weather must comply with recommendations of ACI 306R. All concrete must be moist cured for a minimum of 7 days in accordance with specification requirements.

Exterior concrete must be air entrained with the total air content measured at the point of placement in compliance with ACI 318. Concrete slump must be in accordance with requirements in the concrete specification. Aggregate must meet minimum quality requirements of ASTM C33 and must comply with the appropriate Class Designation in ASTM C33.

4.9. MASONRY

Masonry must be designed as reinforced but must NOT be designed by the empirical method. Shear walls must be running bond construction only; stack bond construction is not permitted. Horizontal wall reinforcement must be continuous around wall corners and through wall intersections, unless the intersecting walls are separated. Reinforcement that is spliced in accordance with the applicable provisions of ACI 530 must be considered continuous. Masonry walls below grade must be grouted solid. Spacing and placement of control joints must be in accordance with NCMA TEK 10-03.

Masonry materials must conform to the following requirements:

Concrete Masonry: $f'_m = 1,500$ psi.
Concrete Masonry Units: ASTM C90, normal weight
Type S mortar
Grout: $f'_c = 2000$ psi at 28 days.

4.10. STRUCTURAL STEEL

Structural steel must be designed and constructed in accordance with the provisions of AISC 325 and AISC 360. Design of cold-formed steel structural members must be in accordance with AISI S100. Welds must be in accordance with AWS D1.1/D1.1M.

4.10.1. Reinforcing Steel

Reinforcing steel must be deformed bars conforming to American Society for Testing and Materials (ASTM) publication ASTM A615/A615M, grade 60, Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement. Concrete members at or below grade must have a minimum concrete cover over reinforcement of 3 inches.

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5. MECHANICAL

5.1. GENERAL

The mechanical design shall be based upon the requirements provided below. The Contractor shall design and construct all mechanical components for the project. This includes design, construction, all necessary labor, equipment, and material for a fully functional mechanical system required for satisfactory operation of the projects. The design of the systems shall be in accordance with the requirements contained herein and the latest edition of UFC 3-460-01 as referenced. The design and product requirements contained herein are minimum requirements. The design and selection of materials and equipment shall be submitted to the Contracting Officer for approval. The Contractor should consider a construction plan that minimizes or eliminates infringement upon normal daily operations. The Contractor shall perform a site survey prior to design start and is responsible for obtaining all necessary information and performing all calculations, measurements, and any other data pertinent to design and construction of this project.

5.2. EXISTING SYSTEM DESCRIPTION AND SCOPE OF WORK

EXISTING SYSTEMS:

The existing systems serving the Fuel Dispatch Building and the POL Admin & Operations building is outdated and in need of replacement. The existing systems will be removed in their entirety and replaced with new mechanical systems.

The existing systems that serve the laboratory building are not adequate for the function of the building and should be removed and replaced with new mechanical systems to create a safe operating environment.

NEW HVAC AND PLUMBING DESCRIPTION:

Laboratory 9647:

The HVAC system for the laboratory portion of the building shall include a make-up air unit capable of fully cooling, dehumidifying, heating, and reheating all of the required outside air to maintain the space at a slightly negative pressure relative to the office. Exhaust from the laboratory shall utilize a high plume dilution fan and be capable of operating to maintain space pressure as described above.

The laboratory office shall utilize a small split system with required ventilation to maintain the space positive relative to the laboratory.

Plumbing work shall include sanitary and domestic water piping for the new double bowl sink, emergency shower/eyewash and fume hoods as required. A new electric domestic water heater shall be installed to ensure that the required water temperature can be delivered to the emergency shower for a minimum of 20 minutes.

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5.3. DESIGN CRITERIA AND REFERENCE PUBLICATIONS

AMERICAN SOCIETY OF HEATING, REFRIGERATION, AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE Handbooks Latest edition

ASHRAE 15 Safety Standard for Refrigeration Systems (2016)

ASHRAE 34 Designation and Classification of Refrigerants (2016)

ASHRAE 62.1 Ventilation for Acceptable Indoor Air Quality (2016)

ASHRAE 90.1 Energy Standard for Buildings Except Low-Rise Residential Buildings (2019)

ASHRAE 189.1 International Green Construction Standard (2014)

INTERNATIONAL CODE COUNCIL (ICC)

IPC International Plumbing Code (2022)

IMC International Mechanical Code (2022)

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

ANSI/ISEA Z358.1 American National Standard for Emergency Eyewash and Shower Equipment (2014)

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA 30 Flammable and Combustible Liquids Code (2021)

NFPA 30A Code for Motor Fuel Dispensing Facilities and Repair Garages (2021)

NFPA 101 Life Safety Code (2021)

NFPA 704 Standard System for the Identification of the Hazards of Materials for Emergency Response (2022)

NFPA 54 National Fuel Gas Code (2021)

NFPA 90A Standard for Installation of Air-Conditioning and Ventilating Systems (2021)

UNIFIED FACILITIES CRITERIA (UFC)

UFC 1-200-01 DoD Building Code (Nov 2023)

UFC 1-200-02 High Performance & Sustainable Building Reqs (Sept 2022)

UFC 3-410-01 Heating, Ventilating, and Air Conditioning Systems (Nov 2021)

UFC 3-420-01 Plumbing Systems with Change 10 (Oct 2021)

UFC 3-430-01FA Heating and Cooling Distribution Systems (2003)

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UFC 3-460-01 Design: Petroleum Fuel Facilities, (2016; Chg 2 - May 2018)
UFC 3-460-03 Operation and Maintenance of Petroleum Systems (2017)
UFC 3-600-01 Fire Protection Engineering for Facilities (May 2021)
UFC 4-010-01 DoD Minimum Antiterrorism Standards for Buildings (Dec 2018)
UFC 4-010-06 Cybersecurity of Facility-Related Control Systems (Jan 2017)

U.S. DEPARTMENT OF DEFENSE

MIL-STD-161H Identification Methods for Bulk Petroleum Products Systems
Including Hydrocarbon Missile Fuels (2015)

5.3.1. Specification References (Unedited, provided as guidance)

The list below is not an all-inclusive list of all specifications that may be relevant for use during this project. A complete list of available specifications and the specifications themselves are available electronically at: http://www.wbdg.org/ccb/browse_org.php?o=70

UNIFIED FACILITIES GUIDE SPECIFICATIONS (UFGS)

02 41 00	Demolition
05 50 13	Miscellaneous Metal Fabrications
11 53 00	Laboratory Equipment and Fumehoods
22 00 00	Plumbing, General Purpose
23 00 00	Air Supply, Distribution, Ventilation, and Exhaust Systems
23 03 00.00 20	Basic Mechanical Materials and Methods
23 05 15	Common Piping for HVAC
23 05 93	Testing, Adjusting and Balancing for HVAC
23 07 00	Thermal Insulation for Mechanical System
23 09 00	Instrumentation and Control for HVAC
23 09 23.02	BACNET Direct Digital Control for HVAC and Other Building Control Systems
23 23 00	Refrigerant Piping
23 31 13.00 40	Metal Ducts
23 37 13.00 40	Diffusers, Registers, and Grills [sic]
23 81 00.00 20	Unitary Air Conditioning Equipment
23 82 16.00 40	Air Coils

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23 82 23.00 40 Fan Coil Units

23 82 46.00 40 Electric Unit Heaters

5.4. MATERIALS

Material and equipment shall be a standard product of a manufacturer regularly engaged in the manufacture of the product and shall essentially duplicate items that have been in satisfactory use for at least 2 years. Equipment installed shall be capable of being serviced by an organization that is, in the opinion of the Contracting Officer, reasonably convenient to the site. Materials and equipment shall be installed in accordance with recommendations of the manufacturer. Major components of equipment shall have the manufacturer's name, address, type or style, and catalog number on a non-corrosive and non-heat sensitive plate, securely attached to the equipment. This plate shall remain visible. All equipment delivered and placed in storage prior to installation shall be protected from the weather, humidity and temperature variation, dirt and dust, and any other contaminants. All major equipment shall have equipment tags or identification plates. Carbon steel flanges shall use carbon steel nuts and bolts. Stainless steel and SS to CS flanges shall use stainless steel nuts and bolts.

5.5. SUSTAINABILITY AND ENERGY EFFICIENCY POLICY

See paragraph 2.1.1 Sustainability and Energy Efficiency Policy above.

5.6. PHASING

See phasing requirements provided in section 1.2 GENERAL.

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6. ELECTRICAL

6.1. SCOPE OF WORK

The Contractor shall finish the design, provide, construct and test all electrical components for the project as listed below. This includes design, testing, construction, necessary labor, equipment, and material for a fully functional electrical system required for satisfactory operation of the work and its interaction and connection to the existing system.

The design and construction shall be in accordance with the latest edition of UFC 3-460-01 as well as applicable codes, standards and other reference criteria given herein. Equipment not covered by other standards shall comply with the latest edition of the National Fire Protection Association (NFPA) Standard 70, National Electrical Code.

The design and construction of the electrical system shall include, but not be limited to:

Fuel Laboratory Facility 9647 (CLIN 004)

1. Demolition and construction for the complete renovation of the building.

6.2. GENERAL REQUIREMENTS

6.2.1. Design Criteria, Applicable Standards and References

The complete design and installation of the components of the electrical system shall be in accordance with the requirements of the standards, Unified Facilities Criteria (UFC), Design Guides and references given below. The latest edition available at the time of Contract Award shall apply. The standards, UFC, Design guides and references given below are not intended to limit the design criteria nor construction methods nor to omit other applicable methods and guidance set forth in any other material not in direct conflict with the publications given below. In every instance where the criteria are applicable, they shall become part of this contract document. In the event of a conflict between the UFC and the contract, the contract shall be followed. The following are to be included, but are not limited to:

American Petroleum Institute (API)

API RP 500 - Classification of Locations for Electrical Installations at Petroleum Facilities Classified as Class 1, Division 01 and Division 02, 2012
API RP 2003 - Protection Against Ignitions Arising out of Static, Lightning, and Stray Current - Eighth Edition, 2015

American Society of Heating, Refrigeration and Air-Conditioning Engineers (ASHRAE)

ANSI/ASHRAE/IES Standard 90.1-2013

Sustainability and Energy Efficiency Policy - see paragraph 1.2.2 above

Illuminating Engineering Society (IES)

The Lighting Handbook, 10th Edition

International Building Code (IBC)

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IBC 2021

Institute of Electrical and Electronics Engineers (IEEE)

IEEE C2 (2023) National Electrical Safety Code

National Association of Corrosion Engineers (NACE)

NACE SP0169 Control of External Corrosion on Underground or Submerged Metallic Piping Systems
NACE SP0177 Mitigation of Alternating Current and Lightning Effects on Metallic Structures and Corrosion Control Systems
NACE SP0188 Discontinuity (Holiday) Testing of New Protective Coatings on Conductive Substrates
NACE RP0286 Electrical Isolation of Cathodically Protected Pipelines.

National Fire Protection Association (NFPA)

NFPA 30 - Flammable and Combustible Liquids Code, 2021
NFPA 45 - Standard on Fire Protection for Laboratories Using Chemicals, 2019
NFPA 70 - National Electrical Code (NEC), 2023
NFPA 70E - Standard for Electrical Safety in the Workplace, 2021
NFPA 72 - National Fire Alarm and Signaling Code, 2022
NFPA 77 - Recommended Practice on Static Electricity, 2019
NFPA 101 - Life Safety Code, 2021
NFPA 110 - Standard for Emergency and Standby Power Systems, 2022
NFPA 497 - Recommended Practice for the Classification of Flammable Liquids, Gases, or Vapors and of Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas, 2021
NFPA 780 - Standard for the Installation of Lightning Protection Systems, 2023

Unified Facilities Criteria (UFC)

1-200-02 High Performance & Sustainable Building Reqs (Sept 2022)
3-460-01 Design: Petroleum Fuel Facilities, With Change 3 (June 2023)
3-460-03 O&M: Maintenance Of Petroleum Systems, With Change 2 (Aug 2023)
3-501-01 Electrical Engineering, With Change 1 (Nov 2019)
3-520-01 Interior Electrical Systems, With Change 2 (Apr 2021)
3-530-01 Interior And Exterior Lighting Systems (Feb 2023)
3-550-01 Exterior Electrical Power Distribution, With Change 3 (Nov 2019)
3-575-01 Lightning And Static Electricity Protection Systems, With Change 1 (Oct 2021)
3-580-01 Telecommunications Building Cabling Systems Planning and Design, with Chg 1, 06-01-2016
3-600-01 Fire Protection Engineering For Facilities, With Change 6 (May 2021)
4-310-03 DoD Fuels Laboratory Standards (12 Aug 2020)

Unified Facilities Guide Specifications (UFGS)

26 00 00.00 20 Basic Electrical Materials and Methods
26 05 48.00 10 Seismic Protection for Electrical Equipment
26 20 00 Interior Distribution System
26 28 00.00 10 Motor Control Centers, Switchboards and Panelboards
26 28 01.00 10 Coordinated Power System Protection
26 41 00 Lightning Protection System

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26 51 00	Interior Lighting
26 56 00	Exterior Lighting
27 10 00	Building Telecommunications Cabling Systems
28 31 34.00 10	Fire Detection and Alarm System, Addressable
33 08 55	Commissioning of Fuel Facility Systems
33 09 53	Aviation Fuel Pump Control and Annunciation System
33 71 02	Underground Electrical Distribution
33 82 00	Telecommunications Outside Plant (OSP)

6.2.2. Site Survey

The Contractor shall perform a site survey prior to design start and is responsible for obtaining all necessary information and performing all calculations, measurements, and any other data pertinent to design and construction of this project.

The Contractor's field survey shall be performed to gather information for this design, field verify any potential problems, and report and make recommendations to the COR for proposed resolutions. The Contractor shall note all existing utilities, structures and enclosures that could possibly impact the electrical design and power supply for this project and provide a written report to the COR. The report shall include findings and recommendations on how to utilize these existing facilities if possible or to integrate or utilize new facilities into the final design.

6.2.3. Demolition of Existing Electrical Work

The Contractor shall perform demolition required to prepare the facilities for the new work or as directed by the description of the work to be performed. All fixtures, material and equipment removed as a result of demolition are considered Government property and shall be properly stored at a location approved by the COR or disposed of off Government property as directed by the COR. No demolition shall take place prior to obtaining authorization to do so.

6.3. MATERIALS

6.3.1. General

The design and selection of materials and equipment shall be submitted to the Contracting Officer for prior approval by the end of the design process. Unless noted otherwise, all material used shall be in compliance with ASTM Standards or Underwriters Laboratories Inc. (UL) listed material. Material and equipment shall be a standard product of a manufacturer regularly engaged in the manufacture of the product and shall essentially duplicate items that have been in satisfactory use for at least 2 years. Materials and equipment installed shall be capable of being serviced by an organization that is, in the opinion of the Contracting Officer, reasonably convenient to the site. Materials and equipment shall be installed in accordance with manufacturer's recommendations. Major components of equipment shall have the manufacturer's name, address, type or style, voltage and current rating, and catalog number on a non-corrosive and non-heat sensitive plate, securely attached to the equipment.

6.3.2. Storage

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All equipment delivered and placed in storage, prior to installation, shall be protected from the weather, humidity and temperature variation, dirt and dust, and any other contaminants. Failure to protect the electrical components and maintain them in a condition acceptable to the manufacturer or contracting officer shall be replaced with new from the same manufacturer at the option of the government. The government shall not be held liable for the charges associated with the procurement and shipping of the replacement electrical components nor charges associated with the removal of the unacceptable electrical components.

6.4. ELECTRICAL DESIGN

6.4.1. General

The electrical design documents included in this RFP package include this SOW. The Contractor shall be responsible for preparing complete design documents to the 100% level. The design effort shall include but not be limited to the following: hazardous area classification information, lightning protection system calculations, electrical calculations, lighting calculations, material submittals, finished contract drawings, and finished specifications.

Power calculations shall confirm existing electrical service is adequate for new loads via new estimated demand load. Load calculations shall be in accordance with NFPA 70. If the addition of the new load exceeds the existing service capacity, contact the contracting officer. Voltage drop and short circuit calculations shall be provided as required. Arc-flash hazard warning shall be provided in accordance with section 110.16 of NFPA 70.

Lighting calculations shall be provided with target illuminance, calculated max, min and average illuminance, target uniformity and calculated uniformity. Lighting design shall meet IESNA Lighting Handbook and API RP 540.

6.4.2. Design Supporting Data

Contractor shall furnish copies of all appropriate sections from pertinent standards, i.e. API and catalog cuts from manufacturers, as back-up material to support Contractor's design of all major components listed below. Referencing appropriate sections from applicable standards in a narrative or in the design analysis is not acceptable. Any variation to, or deviation from, the standards which are to be followed in the development of this design, shall be submitted in accordance with Section 01 33 00 entitled SUBMITTAL PROCEDURES.

6.5. BUILDING REQUIREMENTS

Electrical utility service is base owned. All exterior work shall be coordinated with base CE. All exterior communications work will be coordinated with Base Comm Squadron.

6.5.1. Building: Construct/Renovate Fuel Laboratory Facility 9647 (CLIN 0001)

- Demolition of existing electrical service and distribution back to service point at the meter, on the north face of the building.

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- The electrical loads in the building are not expected to increase significantly. The service feeding the building should not require replacement.
- Demolition of branch wiring, devices, and lighting.
- Existing exterior lighting is surface mounted HID luminaires. These will be removed and replaced with LED sources.
- Demolition of telecom devices and cabling back to service point at pedestal north of the building.
- New electrical distribution.
- Branch wiring for devices and lighting.
- New telecom cabling and devices.
- Addition of a fire alarm panel, to monitor the new sprinkler system. The system will not require mass notification
- Interior lighting will utilize LED sources. Emergency egress lighting will be wall/ceiling mounted unit equipment with LED sources. Exit signs will be LED. Interior lighting will utilize LED sources. Emergency egress lighting will be wall/ceiling mounted unit equipment with LED sources. Exit signs will be LED.
- The current building does not have CATV, CCTV, or IDS systems. These systems will not be added to the renovation.

6.6. CABLES, INSTALLATION AND CIRCUITING

All wiring and cables shall be installed in conduits. Minimum wire size, except for control cable, shall be 12 AWG. Interior wiring and cables shall be placed in surface mounted EMT conduits. Underground cables shall be placed in PVC (Schedule 40, min.) conduits except where restricted by NEC. Exterior cables shall be placed in rigid metal conduits. Cable warning tape shall be provided 6 inches above all underground cables. All splicing and terminations of wires shall be performed in a junction or device boxes. Proper wire nuts/connectors shall be used for splicing wire. No twist-wire connections with electrical tape wrapped around it shall be acceptable. All electrical installation shall be in accordance with the applicable requirements of NFPA 70. Contractor shall be responsible for providing hand-holes, manholes and/or pull boxes, wherever deemed necessary.

6.7. SPECIAL REQUIREMENT

All electrical equipment, fixtures and components used and installed within hazardous areas as defined by the NFPA 70 and UFC 3-460-01 shall be suitable for those locations.

6.8. ON SITE TRAINING

The contractor shall furnish the services of competent instructors to give full instruction to designated Government personnel in the adjustment, operation, and maintenance of the electrical systems and equipment, including pertinent safety requirements as required. Instructors shall be thoroughly familiar with all parts of the installation and shall be trained in operating theory as well as practical operation and maintenance work. Instruction shall be given during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. The contractor shall provide one (1) man-day (8 hours) of instruction. Training shall include a minimum of two (2) hours of classroom training. All training shall be recorded and provided to Base personnel for future use. When

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significant changes or modifications in the equipment or system are made under the terms of the contract, provide additional instructions to acquaint the operating personnel with the changes or modifications.

6.9. ACCEPTANCE TESTING

All systems shall be tested in the presence of the Contracting Officer, or their designated representative, for satisfactory operation prior to acceptance.

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7. FIRE PROTECTION REQUIREMENTS

7.1. REFERENCES

The publications are referred to in the text by the basic designation only. The most current edition shall be used whenever a specific edition is not mentioned.

ASTM INTERNATIONAL (ASTM)

ASTM E84 (2015b) Standard Test Method for Surface Burning Characteristics of Building Materials

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC (2024) International Building Code

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA) - Use most recent editions

NFPA 1 Fire Code

NFPA 10 Standard for Portable Fire Extinguishers

NFPA 13 Standard for the Installation of Sprinkler Systems

NFPA 20 Standard for the Installation of Stationary Pumps for Fire Protection

NFPA 24 Standard for the Installation of Private Fire Service Mains and Their Appurtenances

NFPA 30 Flammable and Combustible Liquids Code

NFPA 45 Standard on Fire Protection for Laboratories Using Chemicals

NFPA 54 ANSI Z223.1 - National Fuel Gas Code

NEC 70 National Electrical Code

NFPA 72 National Fire Alarm and Signaling Code

NFPA 90A Standard for the Installation of Air-Conditioning and Ventilating Systems

NFPA 101 Life Safety Code

NFPA 170 Standard for Fire Safety and Emergency Symbols

NFPA 220 Standard on Types of Building Construction

NFPA 221 Standard for High Challenge Fire Walls, Fire Walls, and Fire Barrier Walls

NFPA 241 Standard for Safeguarding Construction, Alteration, and Demolition Operations

NFPA 291 Recommended Practice for Fire Flow Testing and Marking of Hydrants

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NFPA 5000 Building Construction and Safety Code

U.S. DEPARTMENT OF DEFENSE (DOD) - Use most recent editions

UFC 1-200-01 DoD Building Code

UFC 3-301-01 Structural Engineering

UFC 3-230-01 Water Storage and Distribution

UFC 3-600-01 Fire Protection Engineering for Facilities

UFC 4-021-01 Design and O&M: Mass Notification Systems

UFC 4-021-02 Electronic Security Systems

UFC 4-310-03 DoD Fuels Laboratory Standards

Refer to the Architectural section for a description of the work required for this project. The following is a summary of the work:

Building 9744 Crew Fuel Dispatch - Renovate Existing Building
Building 9647 Fuels Lab - Renovate Existing Building
Building 9691 Pump Crew Shelter - New Building
Building 9941 Pump Crew Shelter - New Building
Building 7229 POL Admin and Operations Facility - Renovate Existing Building
Building 7931 Restroom at Pump House - Renovate Existing Building.

A life safety and building code plan shall be developed for each of the facilities. Egress and building components shall meet NFPA 101 criteria.

The fuels lab (Building 9647) shall be provided with a fire sprinkler system. Provide a new fire hydrant flow test per NFPA 291 prior to starting the design of the sprinkler system.

A new fire alarm and mass notification system shall be provided for each facility. Egress and building components shall meet NFPA 101 criteria.

Fire protection shall be based on sound fire protection engineering principles and shall give safeguards against loss of life and property by fire, consistent with the mission, risk involved, and economical utilization. Fire protection criteria shall also be based on the codes listed above and on the following code requirements:

ABA Accessibility Guidelines for Buildings and Facilities
(www.access-board.gov/ada-aba/final.cfm)

All requirements of the latest codes and criteria references noted before shall be incorporated into the design. A Fire Protection Engineer shall be part of the design team in accordance with UFC 3-600-01. The Fire Protection Engineer shall provide a Life Safety and Building Code Analysis as well as be responsible for the design of the fire alarm, fire sprinkler, and mass notification systems.

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The Life Safety Code

NFPA 101 relative to this design shall give special attention to the application of fire codes as they relate to Life Safety. Features of fire protection based on the following shall be included in the design: Automatic operating devices; Exiting for inhabitants and the protection of egress components; Personnel safety in hazardous areas; Appropriate ratings of fire and smoke partitions, doors and windows; Travel distances; Common paths of travel; Occupancy types; Hazard of occupancies and their contents; and Isolation from the remainder of the facility.

Applicable requirements of the International Building Code shall also be included in the design. These shall include the following: Types of construction; Fire area limitations; Increases to allowable floor areas; and Separation of structures.

All military construction must comply with the code requirements set forth in the UFC criteria mentioned before such as UFC 1-200-01, UFC 3-600-01, UFC 4-021-01, and UFC 4-310-03.

All fire protection design shall be done by a Fire Protection Engineer in accordance with UFC 3-600-01. The fire protection engineer shall perform a life safety and building code analysis of the renovated facility including egress capacity and pathways. The life safety and building code analysis shall be submitted with the first submittal for review. Life safety analysis shall comply with UFC 3-600-01 and NFPA 101. The new floor plans shall be evaluated for compliance with NFPA 101.

7.2. OCCUPANCY FOR THE PROJECT

The facilities will be classified as B Business in accordance with NFPA 101. According to Chapter 4 of the International Building Code (IBC), the project will be classified as "Group B" (Business) Occupancy.

7.3. CONSTRUCTION PER 2024 INTERNATIONAL BUILDING CODE (IBC)

Minimum Type II-B, except that hourly fire rating requirements for the building systems and components shall not be less than those specified in the performance specifications sections.

7.4. CORRIDORS

Review and provide in compliance with NFPA 101.

Separation of incidental use areas shall be provided per NFPA 101.

7.5. INTERIOR FINISHES

Interior wall and ceilings of exits shall have Class A or Class B finishes in accordance with NFPA 101.

Interior floor finishes in exits shall be Class I or Class II in accordance with NFPA 101.

No downgrade in finish Class due to complete coverage by an automatic sprinkler system shall be allowed.

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Smoke Developed Ratings and Flame Ratings shall be in accordance with NFPA 101.

Provide self-closing hardware and gasketing at all fire-rated doors.

7.6. EGRESS CAPACITY:

Egress capacities shall be calculated and provided in compliance with NFPA 101.

7.7. MEASUREMENT OF TRAVEL DISTANCE TO EXITS:

Review and provide in compliance with NFPA 101

7.8. FIRE EXTINGUISHER CABINETS:

Provide per NFPA 101, NFPA 10, and UFC 3-600-01.

7.9. AUTOMATIC SPRINKLER SYSTEMS:

The lab facility shall be provided with a fire sprinkler system and meet UFC 3-600-01. New underground water line, riser, valves, fire department connection, backflow preventer, forward flow test connection, flow switch, and tamper switches shall be required for the building. System shall be designed per UFC 3-600-01 and installed per NFPA 13. Provide a new hydrant flow test in accordance with NFPA 291 and use that data to design the system. Areas and densities shall comply with UFC 3-600-01. Hydraulically calculate the new system. For the new water line, the maximum velocity in the underground pipe shall be 10 fps. System is to be a wet pipe sprinkler system. Lab areas shall be considered Ordinary Hazard. Storage rooms shall be classified as Ordinary Hazard.

7.10. PRESENCE OF DISABLED OCCUPANTS

Disabled personnel will potentially be present. Provisions for accessibility and usability will be made for physically handicapped individuals for exiting these facilities.

7.11. FUNCTIONAL AND TECHNICAL REQUIREMENTS

7.11.1. Building Construction Type

These facilities shall comply with a minimum Construction Type II-B, in accordance with ICC IBC.

7.11.1.1. Exterior Walls

Exterior walls of the facilities will not be rated if minimum distances from other buildings are maintained.

7.11.1.2. Roof

The building roof covering shall be Factory Mutual Approved or classified by Underwriter's Laboratory as a minimum Class B roof system.

7.11.1.3. Interior Walls

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All penetrations in fire and smoke rated walls (conduits, pipes, cable trays, etc.) shall be fire or smoke stopped according to their respective wall/floor/ceiling rating at each penetration.

7.12. FIRE ALARM AND DETECTION SYSTEMS

7.12.1. Fire Alarm Panel

Each building shall be provided with a new addressable fire alarm panel (FACP). All new circuits to the new addressable type fire alarm systems shall comply with NFPA 72, UFC 3-600-01, and NFPA 101. Contractor shall install new detectors and initiating devices as required for the facility. Provide a new fire alarm and mass notification system in compliance with NFPA 72, NFPA 101, UFC 3-600-01, UFC 4-021-01 and ABA.

The system shall be addressable to each reporting device and with turnkey MNS. MNS devices shall be combination speaker/strobe on ceiling. The system shall be complete with the control panel in the building and required devices. The addressable system shall transmit and receive addresses and data between the control panel and the new devices. Provide LOCs in accordance with UFC 4-021-01.

The fire alarm system shall have Class A, Style 6 ALPHA signaling line circuits, Class A, Style E ALPHA for initiating device circuits, and Class A, Style Z for notification appliance circuits.

7.12.2. Initiating and Notification Devices

The new systems shall include addressable manual pull stations, addressable heat detectors, addressable duct smoke detectors, addressable spot type smoke detectors, and audible and visual notification appliances that comply with ABA requirements. The designs shall include all fire sprinkler devices such as flow, tamper, pressure, etc. and supervisory switches. These devices shall be connected to the control panel through an addressable interface module. Provide a waterproof strobe and speaker at the location of the fire department connection.

7.12.3. Mass Notification System

Refer to UFGS Section 28 31 76 INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM and UFC 4-021-01 for requirements for the Mass Notification System. Audibility and Intelligibility shall follow UFC 4-021-01.

7.12.4. Interfaces to Other Systems

The new system shall shut down appropriate air handling equipment, smoke dampers, and release magnetic door holding devices as necessary for proper operation of the facilities. Provide a "Sequence of Operations Matrix" to coordinate with alarm functions. The "Sequence of Operations Matrix" shall be approved by the Contracting Officer. System shall meet requirements of UFC 4-021-02.

7.12.5. Layout Considerations

Duct smoke detectors shall be provided on all new air handling supply fans over 2000 CFM, and all new return fans over 15,000 CFM. Smoke detectors

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shall be provided where electromagnetic door holders are used. Audible and visual notification appliances shall be provided to meet these specifications which includes the new interior egress areas. Design shall comply with ABA Requirements, NFPA 101 NFPA 72 and UFCs. Provide a waterproof audible notification appliance on the exterior at each exit door. Addressable initiating device circuit modules shall be provided for each non-addressable device, excluding notification appliances.

7.13. FIRE PROTECTION SPECIFICATIONS (SUBMITTALS AND ACCEPTANCE TESTS)

USACE is the Authority Having Jurisdiction for all fire protection systems.

The sprinkler and fire alarm submittals shall be reviewed and approved by the Base Fire Chief or designated representatives as well as USACE Omaha District Fire Protection Engineer.

Representatives of the Base Fire Department shall inspect and witness all final acceptance testing prior to the Contracting Officer accepting the sprinkler and fire alarm systems.

Fire water service shall be tested in accordance with NFPA 24 guidelines. Fire sprinkler system shall be tested in accordance with NFPA 13 guidelines. Fire alarm and mass notification shall be tested in accordance with NFPA 72 guidelines.

7.14. DESIGN OBJECTIVES AND PROVISIONS

7.14.1. Zoning and Treatment of Each Potential Hazard

7.14.1.1. Limiting Fire Spread

Every horizontal opening and hazardous locations as defined by NFPA 101 shall be protected.

7.14.2. Provision and Maintenance of an Unobstructed Emergency Egress System

All corridor widths, clear space requirements relative to exit doors, etc., shall be in accordance with the Americans with Disabilities Act for unobstructed egress. Emergency lighting shall be installed in accordance with NFPA 101.

7.14.3. Maximum dead ends.

Maximum dead ends shall be as per NFPA 101.

7.14.4. Egress locations

Egress locations shall be marked with exit signs per NFPA 101.

7.14.5. Outside Exit Doors

Outside exit doors shall swing in the direction of exit travel. Outside exit doors shall be equipped with panic hardware mounted 44 inches above the finish floor and have a minimum clear width of 34 inches to allow for egress.

7.14.6. Required Fire Exits

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Required fire exits from the building shall be provided in compliance with NFPA 101 and lead to a public way or to a clear safe area at a minimum distance of 75-feet from the building.